

Boundedly Rational Trust

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Modeling Question:

When can humans justifiably trust AI?
(or can't?)

- Can we prompt our way to Paradise?
- Is there a **basin of attraction** for alignment or corrigibility?
 - ↳ What determines whether we're in or out?

Deference Done Better by

Kevin Dorst
Ben A. Levinstein
Bernhard Salow
Brooke E. Husic
Branden Fitelson

VALUE : Novice is happy ($\geq$) to defer any decision to the expert.

IMMODESTY: The expert doesn't know their own beliefs perfectly.

Earlier result (Skyrms 1990): supposing IMMODESTY,

$$\underbrace{P_{\text{nov}}(X | P_{\text{exp}}(X) = x)}_{\text{REFLECTION}} = x \iff \text{VALUE}$$

Deference Done Better result:

$$\underbrace{E_{\text{nov}}(V | E_{\text{exp}}(V) \geq v)}_{\text{TOTAL TRUST}} \geq v \iff \text{VALUE}$$

No IMMODESTY needed!

Introspection/Modesty discussion:

- all that's fair under introspection
- Dorst

Reflection $\Leftrightarrow$ VALUE $\Leftrightarrow$ Total Trust $\Leftrightarrow$ Tower
will come
up later

A Decision-Theoretic Model for Managing Misalignment

by Daniel A. Hermann, Abinav Chari, Isabelle Qian, Sree Sharvesh, & Ben A. Levinstein
applies the model to AI alignment, studying differences in utility functions amongst other results.

This seems like a productive model.

However,

→ It is not yet very constructive: how do we build something aligned in this sense?

→ The authors of Deference Done Better complain that the proof is horrible

→ Deference and Infinite Frames by Brian Weatherston

shows that the result is quite sensitive to assumptions, which calls the practical implications into question

→ Can we "verify" the idea of DDB in another setting?

Logical Induction by Scott Garrabrant, Tsvi Benson-Tilsen,
Andrew Critch,
Nate Soares,
Jessica Taylor

→ Epistemic processes rather than
static belief snapshots

→ Boundedly rational: LI Criterion (LIC) requires
bounded loss against **computationally efficient**
Dutch Books

→ Constructive: LI Algorithm (LIA) is computable

☆ Better model of human scientific & philosophical
progress

☆ Better model of AI incremental releases & RSI

Can we translate DDB to LI?

Translating VALUE:

$$x_n \lesseqgtr y_n := \limsup_{n \rightarrow \infty} (x_n - y_n) \leq 0$$

The random variable
which takes the value
of whichever option
the AI chooses
(argmax with tiebreak)

$$\max_{a \in A_n} \mathbb{E}_n^H(a) \lesseqgtr_n \mathbb{E}_n^H \left(\overbrace{\text{choose}_{a \in A_n} [\mathbb{E}_n^A(a)]} \right)$$

Human expectation
on day n

AI expectation
on day n

A_n any "regular" sequence of decision problems,
expressed as e.d. sets of bounded real variables (LUVs)

Translating TOTAL TRUST (TT):

Original:

multiply →

$$\mathbb{E}_{\text{nov}}(V \mid \mathbb{E}_{\text{exp}}(V) \geq v) \geq v$$
$$\mathbb{E}_{\text{nov}}(V \cdot \text{Ind}(\mathbb{E}_{\text{exp}}(V) \geq v)) \geq v \cdot \mathbb{P}_{\text{nov}}(\mathbb{E}_{\text{exp}}(V) \geq v)$$

↓ translate

$$\mathbb{E}_n^H(V \cdot \text{Ind}_\delta(\mathbb{E}_n^A(V) > v)) \gtrsim_n v \cdot \mathbb{E}_n^H(\text{Ind}_\delta(\mathbb{E}_n^A(V) > v))$$

Soft indicators: $\text{Ind}_\delta(v > w) := \begin{cases} 0 & v \leq w \\ (v-w)/\delta & w \leq v \leq w+\delta \\ 1 & w+\delta \leq v \end{cases}$

When Does $TT \Rightarrow \text{VALUE}$?

TT implies: $\mathbb{E}_n^A(V_n) \succeq_n v \Rightarrow \mathbb{E}_n^H(V_n) \succeq_n v$ (proof in appendix)

Let $a_n^* := \underset{a \in A_n}{\text{choose}} [\mathbb{E}_n^A(a)]$; taking $V_n = a_n^* - a_n^i$,

$$\mathbb{E}_n^A(a_n^* - a_n^i) \succeq_n 0 \Rightarrow \mathbb{E}_n^H(a_n^* - a_n^i) \succeq_n 0 \Rightarrow \underbrace{\mathbb{E}_n^H a_n^i \preceq_n \mathbb{E}_n^H a_n^*}_{\text{for the right } i, \text{ VALUE!}}$$

So we just need $\mathbb{E}_n^A(a_n^* - a_n^i) \succeq_n 0$, to establish VALUE .

Unfortunately, $\mathbb{E}_n^A(a_n^* - a_n^i) \succeq_n 0$ is not true in general.

Counterexample to $\mathbb{E}_n^A(a_n^* - a_n^i) \gtrsim_n 0$:

Assuming $\mathbb{E}_n^A$ is computable & our language extends PA,

all $a_n^i \in \mathcal{A}_n$ could be $a_n^i := \begin{cases} 0 & a_n^i = \text{choose}_{a' \in \mathcal{A}_n} \mathbb{E}_n^A(a') \\ 1 & \text{otherwise} \end{cases}$

IE: AI choice is always worse. $\leftarrow \mathbb{I}(a_n^i \neq \text{choose}_{a' \in \mathcal{A}_n} \mathbb{E}_n^A(a'))$

We can rule this out by assuming:

REGULARITY:

$$\sum_{a_n^j \in \mathcal{A}_n} \underbrace{P_n^A(a_n^* = a_n^j)}_{\text{probability of action } j} \underbrace{\left[\mathbb{E}_n^A(a_n^j | a_n^* = a_n^j) - \mathbb{E}_n^A(a_n^j) \right]}_{\text{difference between conditioning on being selected vs not}} \underbrace{\approx_n 0}_{\gtrsim_n \text{ is enough}}$$

$$\Rightarrow \underbrace{\sum_{a_n^j \in \mathcal{A}_n} P_n^A(a_n^* = a_n^j) \mathbb{E}_n^A(a_n^j)}_{\text{most mass on near-optima}} \lesssim_n \underbrace{\sum_{a_n^j \in \mathcal{A}_n} \mathbb{E}_n^A(a_n^j \cdot \mathbb{I}(a_n^* = a_n^j))}_{\approx_n \mathbb{E}_n^A(a_n^*)}$$

$$\approx_n \max_{a \in \mathcal{A}_n} \mathbb{E}_n^A(a)$$

$$\approx_n \mathbb{E}_n^A(a_n^*)$$

$\therefore \text{VALUE}$ 

VALUE $\Rightarrow$ TOWER

Suppose VALUE: $\max_{a \in \mathcal{A}_n} \mathbb{E}_n^H(a) \lesssim_n \mathbb{E}_n^H\left(\underset{a \in \mathcal{A}_n}{\text{choose}}[\mathbb{E}_n^A(a)]\right)$, for regular $\mathcal{A}_n$.

We want TOWER: $\mathbb{E}_n^H \mathbb{E}_n^A V_n \approx_n \mathbb{E}_n^H V_n$.

Consider $\mathcal{A}_n = \{V_n - \mathbb{E}_n^A(V_n), -\varepsilon\}$ for rational $\varepsilon > 0$.

$\mathbb{E}_n^A(V_n - \mathbb{E}_n^A(V_n)) \approx_n 0$, so choose $\underset{a \in \mathcal{A}_n}{\mathbb{E}_n^A(a)} \approx_n V_n - \mathbb{E}_n^A(V_n)$;

thus by VALUE, $\mathbb{E}_n^H(-\varepsilon) \lesssim_n \mathbb{E}_n^H(V_n - \mathbb{E}_n^A(V_n))$.

Since $\varepsilon > 0$ was arbitrary, $\mathbb{E}_n^H(V_n) \lesssim_n \mathbb{E}_n^H \mathbb{E}_n^A V_n$.

Considering $\mathcal{A}_n = \{\mathbb{E}_n^A(V_n) - V_n, -\varepsilon\}$ instead, we can similarly prove the other direction. 

TOWER $\Rightarrow$ TOTAL TRUST

full circle

relationship to immodesty

Reflection is different

How LI is "between"
modest & immodest

VALUE $\Rightarrow$ Total Trust

Sam's Conjecture

Let n be a positive integer.

Let d be a positive integer.

Let k be a positive integer.

Let m be a positive integer.

Let p be a prime number.

Let q be a prime number.

Let r be a prime number.

Let s be a prime number.

Let t be a prime number.

Let u be a prime number.

Let v be a prime number.

Let w be a prime number.

Let x be a prime number.

Let y be a prime number.

Let z be a prime number.

Let a be a prime number.

Let b be a prime number.

Let c be a prime number.

Let d be a prime number.

Corrigibility

Legitimacy

Questions?

Appendix.

$$TT \text{ implies } \mathbb{E}_n^A(V_n) \gtrsim_n v \Rightarrow \mathbb{E}_n^H(V_n) \gtrsim_n v$$

Proof. Suppose $\mathbb{E}_n^A(V_n) \gtrsim_n v$.

Consider some $\varepsilon > 2\delta > 0$, and $u := v - \varepsilon$. Define:

$$\omega_n := \text{Ind}_\delta(\mathbb{E}_n^A(V_n) > u)$$

Note that $\omega_n \rightarrow 1$, converging to 1 at some finite N .

$$\underbrace{\mathbb{E}_n^H(V_n)}_{\text{provability induction}} \approx_n \underbrace{\mathbb{E}_n^H(V_n \omega_n)}_{\text{Total Trust}} \gtrsim_n (s - \varepsilon) \underbrace{\mathbb{E}_n^H(\omega_n)}_{\text{provability induction}} \approx_n s - \varepsilon$$